

A Zero-Determinant Right Outer Which Is Not Left Outer

Candidate counterexample for arXiv:1304.0518

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Status. Candidate counterexample, likely valid, pending specialist human review. The construction disproves the proposed left/right equivalence already for a bounded analytic element and, using the standard independence of the Hardy index, for every $1 \leq p \leq \infty$.

1 Source question

Blecher and Labuschagne ask whether the determinant-positive equivalence between left and right outer elements remains true when the Fuglede–Kadison determinant is zero [2, p. 10, after Proposition 2.11]. Their convention, which we use throughout, is

$$h \text{ right outer in } H^p \iff [h\mathcal{A}]_p = H^p, \quad h \text{ left outer in } H^p \iff [\mathcal{A}h]_p = H^p.$$

h and the fact that $h_2 \in [A]_{r.}$) Then

$$\|h_1(ba) - \mathbb{1}\|_p \leq \|h_1(ba) - ha\|_p + \|ha - \mathbb{1}\|_p < \epsilon.$$

Hence h_1 is right outer in H^p . But then h_1 is also right outer in H^q . The alternative claim follows analogously. $\square$

We showed in [6] that if $\Delta(h) > 0$ then h is left outer in H^p if and only if h is right outer in H^p . We do not know if this is true in general if $\Delta(h) = 0$. In the next section we explore a restricted setting where this does hold. In view of the fact that $ab = 1$ if and only if $ba = 1$ in M , we expect that this is likely to hold for all outers, but cannot prove this as yet (see Corollary 4.2 for a partial result in this direction).

Figure 1: The open question in arXiv:1304.0518, PDF page 10.

The exact question is whether, without the hypothesis $\Delta(h) > 0$, right outerness is equivalent to left outerness. The construction below gives a negative answer.

2 Proof intuition

Let v be a nonconstant operator-valued inner function. Right multiplication by v preserves a proper closed inner subspace, so a factorization $h = dv$ is automatically a good candidate for failure of left outerness. If d were bounded below, however, left multiplication by d could not erase the defect space of vH^2 , and h would also fail to be right outer.

The key is to take d bounded and injective but extremely small on a sequence of independent Bernoulli coordinates. The model space $H^2 \ominus vH^2$ is a resolvent-kernel copy of $L^2(\mathcal{D})$. Dividing one of its vectors by d produces a weighted sum of independently shifted copies of d^{-2} . We choose the tail of d^{-2} so heavy that this sum diverges almost surely. Thus no nonzero model-space vector lies in the range of left multiplication by d , and dvH^2 is dense. The same d is nevertheless a full-support diagonal, so $\mathcal{A}d$ is dense and $[\mathcal{A}dv]_2 = H^2v \neq H^2$.

3 Construction and statement

Let

$$(\Omega, \mu) = ((0, 1)^{\mathbb{Z}}, \lambda^{\mathbb{Z}})$$

with Bernoulli shift σ , and put $\mathcal{N} = L^\infty(\Omega)$. Let

$$\mathcal{D} = \mathcal{N} \rtimes_\sigma \mathbb{Z}$$

be the finite crossed-product von Neumann algebra, with canonical trace τ and canonical implementing unitary u . Thus, up to the harmless choice of shift orientation,

$$ufu^* = f \circ \sigma^{-1} \quad (f \in \mathcal{N}).$$

Let

$$\mathcal{M} = L^\infty(\mathbb{T}) \bar{\otimes} \mathcal{D}, \quad \mathcal{A} = H^\infty(\mathbb{T}) \bar{\otimes} \mathcal{D}.$$

Equivalently, $\mathcal{A}$ is the weak-star closure of the analytic $\mathcal{D}$ -valued trigonometric polynomials. This is a finite maximal subdiagonal algebra in $\mathcal{M}$, with diagonal $\mathcal{D}$ and conditional expectation given by the zeroth Fourier coefficient.

Fix $0 < r < 1$ and set $q = r^2$. For $\omega = (\omega_k)_{k \in \mathbb{Z}}$ define

$$d(\omega) = \exp\left(-\frac{1}{2\omega_0}\right), \quad X = d^{-2} = \exp\left(\frac{1}{\omega_0}\right).$$

Then $d \in \mathcal{N}$ is bounded, positive, and has support $\mathbf{1}$, while

$$\tau(\log d) = -\frac{1}{2} \int_0^1 \frac{dt}{t} = -\infty, \quad \Delta(d) = 0.$$

Finally define

$$v(z) = (zu - r\mathbf{1})(\mathbf{1} - rzu)^{-1} = b_r(zu), \quad h = dv.$$

For $|z| = 1$, the operator zu is unitary and the scalar Blaschke transform b_r maps the unit circle to itself. Hence v is a unitary-valued, nonconstant element of $\mathcal{A}$, and $h \in \mathcal{A}$.

Theorem 1. *For the finite maximal subdiagonal algebra $\mathcal{A} \subset \mathcal{M}$ above, the bounded analytic element $h = dv$ satisfies $\Delta(h) = 0$ and*

$$[h\mathcal{A}]_p = H^p \quad \text{but} \quad [\mathcal{A}h]_p \neq H^p \quad (1 \leq p \leq \infty).$$

In particular, a zero-determinant right outer need not be left outer.

4 Proof

We first prove the two claims in H^2 .

4.1 The model space of the inner factor

Identify

$$H^2(\mathcal{A}) = \ell^2(\mathbb{N}_0; L^2(\mathcal{D}))$$

by Fourier coefficients. Let W be left multiplication by zu . It is the pure isometry

$$W(x_0, x_1, x_2, \dots) = (0, ux_0, ux_1, \dots)$$

whose wandering space is the constant copy of $L^2(\mathcal{D})$. Left multiplication by v is $b_r(W)$. The usual one-point Blaschke model-space calculation, included here to fix the module orientation, gives

$$K_v := H^2 \ominus vH^2 = \{k_x : x \in L^2(\mathcal{D})\}, \quad (k_x)_n = \sqrt{1-r^2} r^n u^n x. \quad (1)$$

Indeed,

$$k_x = \sqrt{1-r^2} (\mathbf{1} - rW)^{-1} x, \quad W^* k_x = r k_x,$$

so $b_r(W)^* k_x = 0$. Conversely the Cayley transform identity for b_r maps $\ker W^*$ onto $\ker b_r(W)^*$, giving all of the kernel. Also

$$\|k_x\|_2^2 = (1-r^2) \sum_{n \geq 0} r^{2n} \|u^n x\|_2^2 = \|x\|_2^2,$$

so the parametrization is isometric.

4.2 Right outerness

Because $d = d^*$ is bounded, a vector $y \in H^2$ is orthogonal to $h\mathcal{A}$ if and only if

$$\langle dva, y \rangle = 0 \quad (a \in \mathcal{A}) \iff dy \in H^2 \ominus vH^2 = K_v.$$

Suppose $dy = k_x$. Since d has support $\mathbf{1}$, the Fourier coefficients of y are forced to be

$$y_n = \sqrt{1-r^2} r^n d^{-1} u^n x.$$

Consequently, if $y \in H^2$, then

$$\|y\|_2^2 = (1-r^2) \sum_{n \geq 0} q^n \tau(x^* u^{-n} d^{-2} u^n x). \quad (2)$$

We show that the right-hand side is infinite whenever $x \neq 0$. Put

$$X_n = u^{-n} X u^n \in L^0(\mathcal{N}), \quad S_N = (1-r^2) \sum_{n=0}^N q^n X_n.$$

The X_n are independent copies of X . For $t \geq e$,

$$\mu\{X > t\} = \mu\left\{\omega_0 < \frac{1}{\log t}\right\} = \frac{1}{\log t}.$$

Therefore, for all sufficiently large n ,

$$\sum_n \mu\{X_n > q^{-2n}\} = \sum_n \frac{1}{2n|\log q|} = \infty.$$

The second Borel–Cantelli lemma implies that $X_n > q^{-2n}$ infinitely often almost surely. Along those indices, $q^n X_n > q^{-n}$, and hence

$$S_N(\omega) \longrightarrow \infty \quad \text{for almost every } \omega. \quad (3)$$

Let $E_{\mathcal{N}} : \mathcal{D} \rightarrow \mathcal{N}$ be the canonical trace-preserving conditional expectation. For $x \in L^2(\mathcal{D})$ put

$$g = E_{\mathcal{N}}(xx^*) \in L^1(\mathcal{N})_+.$$

Faithfulness gives $g \neq 0$ if $x \neq 0$. Using truncations for the extended positive operators and then trace cyclicity,

$$(1 - r^2) \sum_{n=0}^N q^n \tau(x^* X_n x) = \tau(S_N x x^*) = \int_{\Omega} S_N g \, d\mu.$$

By (3) and monotone convergence, the last expression tends to ∞ for every $x \neq 0$. Thus (2) cannot be finite unless $x = 0$. Then $dy = 0$, and the full support of d gives $y = 0$.

We have proved $(h\mathcal{A})^\perp = \{0\}$ in H^2 , hence

$$[h\mathcal{A}]_2 = H^2. \quad (4)$$

4.3 Failure of left outerness

Let $e_m = \mathbf{1}_{\{d \geq 1/m\}} \in \mathcal{D}$ and

$$a_m = d^{-1}e_m \in \mathcal{D} \subset \mathcal{A}.$$

Then $a_m d = e_m \rightarrow \mathbf{1}$ in L^2 , so $[\mathcal{A}d]_2 = H^2$. Right multiplication by the unitary-valued analytic function v is an isometry of H^2 and has closed range. Therefore

$$[\mathcal{A}h]_2 = [\mathcal{A}dv]_2 = H^2 v. \quad (5)$$

This range is proper. Indeed, if $\mathbf{1} = fv$ for some $f \in H^2$, then the boundary identity forces $f = v^*$. But v is nonconstant (its first positive Fourier coefficient is $(1 - r^2)u$), so v^* has a nonzero negative Fourier coefficient and does not belong to H^2 . Hence $H^2 v \neq H^2$.

Equations (4) and (5) prove the counterexample for $p = 2$.

4.4 All Hardy indices and the determinant

Blecher and Labuschagne’s index-independence lemma says that, for $1 \leq p \leq \infty$, an element $f \in L^p(\mathcal{M})$ satisfies $[f\mathcal{A}]_p = H^p$ exactly when it is right outer in H^1 [1, Lemma 4.1]; the left-handed statement follows by passing to the opposite algebra. Since our h is bounded, the $p = 2$ conclusions therefore hold for every $1 \leq p \leq \infty$.

Finally v is unitary-valued, so multiplicativity of the Fuglede–Kadison determinant gives

$$\Delta(h) = \Delta(d)\Delta(v) = 0.$$

Also $\Phi(v) = -r\mathbf{1}$ and $\Phi(h) = -rd$, so $\Delta(\Phi(h)) = 0$, as expected for the determinant-zero regime of the source question. This completes the proof of Theorem 1. $\square$

5 Verification notes

1. **Hypotheses.** The algebra $H^\infty(\mathbb{T})\bar{\otimes}\mathcal{D}$ is a standard finite maximal subdiagonal algebra, and $h = dv$ is bounded and analytic.
2. **No hidden bounded-inverse assumption.** The proof uses only that d is bounded and has full support. Its inverse is an affiliated operator; the possible failure of $d^{-1}u^n x$ to lie in L^2 is exactly the mechanism measured by (2).
3. **Left/right orientation.** Right outerness concerns left multiplication by dv on vH^2 and is proved through $H^2 \ominus vH^2$. Failure of left outerness concerns right multiplication by v and yields the distinct subspace H^2v .
4. **Probabilistic step.** The shifted X_n are independent because they are functions of distinct Bernoulli coordinates. The chosen tail makes the Borel–Cantelli sum harmonic.
5. **Extended-positive step.** Every use of d^{-2} can be read with spectral truncations first. Monotone convergence then justifies both trace cyclicity and the expectation identity.
6. **Computational dependence.** None. The construction and proof are exact; no finite experiment or numerical lemma is used.

6 Novelty and limitations

The bounded novelty search performed on 9 August 2026 included:

- the run’s registry, solution, attempt, and proof-gap indexes, plus its local 2000–2026 arXiv source corpus;
- exact and close-variant web/arXiv searches for “left outer”, “right outer”, “determinant zero”, “maximal subdiagonal”, and the title/arXiv id;
- the OpenAlex citation graph for DOI 10.4064/sm217-3-4, containing 37 records. The seven substantive mathematical citing works concerned Orlicz–Hardy spaces, symmetric Hardy spaces, Haagerup Hardy spaces, invariant subspaces, or Beurling-type theorems; none found in the search stated a resolution or counterexample to the exact determinant-zero question.

This bounded search is evidence, not a proof of novelty. The result should be checked by a specialist in noncommutative Hardy spaces, with particular focus on the model-space orientation and the extended-positive conditional- expectation calculation. The packet makes no claim about whether every right outer is uniform outer; it only settles the left/right equivalence negatively.

7 Human-review recommendation

Recommendation: high-priority expert verification. The proof is short enough to audit locally and, if correct, gives a full negative answer to the explicit question in the source paper. A reviewer should first verify (1), then audit the implication $dy = k_x \Rightarrow (2)$, and finally confirm that the crossed-product expectation turns the energy into $\int S_N g$ for arbitrary $x \in L^2(\mathcal{D})$.

References

- [1] D. P. Blecher and L. E. Labuschagne, *Applications of the Fuglede–Kadison determinant: Szegő’s theorem and outers for noncommutative H^p* , Trans. Amer. Math. Soc. **360** (2008), 6131–6147.
- [2] D. P. Blecher and L. E. Labuschagne, *Outers for noncommutative H^p revisited*, Studia Math. **217** (2013), 265–287; arXiv:1304.0518.